

Erdős Problem #611

EP-611

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

For a graph G let $\tau(G)$ denote the minimal number of vertices that include at least one from each maximal clique of G (sometimes called the clique transversal number).

Is it true that if all maximal cliques in G have at least cn vertices then $\tau(G) = o_c(n)$?

Similarly, estimate for $c > 0$ the minimal $k_c(n)$ such that if every maximal clique in G has at least $k_c(n)$ vertices then $\tau(G) < (1 - c)n$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>